VIDEO GAME CREATION

Lesson 14 - Raising the Stakes

Semester 1, Week 14 - Difficulty Scaling & Stretch Features

- **Objectives:** Replace hardcoded per-wave enemy count and speed with formulas keyed off the existing wave variable, keeping every referenced value in agreement; choose and make a good-faith attempt at ONE optional capstone-level stretch feature.
- **IGDA Tie-In:** Game Design - balance, pacing, difficulty curves (intermediate); Programming - parameterization, avoiding duplicated logic (intermediate)
- **Featured Engine (lecture demo):** GameMaker (GML text/script - second script-mode rotation slot, callback to Lesson 9)
- **Materials:** Demo laptop with all 8 engines verified; Student Handouts, one per engine; students' own Lesson 13 projects (already saved); no new assets - 001 Game Creator students add 2 pre-placed enemy copies at design time

Agenda

0:00-0:05	Welcome back, Lesson 13 recap
0:05-0:15	The Foreshadowed Feature: One Formula, Two Places
0:15-0:35	Adding Speed to the Mix (live demo in GameMaker)
0:35-0:45	This Week's Twist: Student's Choice
0:45-0:55	Introduce lab exercise, hand out engine-specific handouts
0:55-1:50	LAB: Scale count and speed, then pick one stretch feature
1:50-2:00	Exit check / quick share-out - toughest wave 3 wins bragging rights

Levels for this lesson's exercise

- Level 1 (Minimal): enemy count increases by one each wave (5, 6, 7), using a single formula referenced everywhere it's needed.
- Level 2 (Complete): enemy speed ALSO increases each wave; restarting correctly returns everything to wave 1's original values.
- Level 3 (Refined): choose exactly ONE of five capstone-level stretch features (shields, power-up, bonus enemy, formation movement, title/game-over screen) and make a good-faith attempt.

See: Lesson 14 Teacher's Script, Lesson 14 Prep Checklist, and the per-engine Student Handouts. Level 3 this week is open-ended student choice - expect a wider range of finishing points than usual, and that's fine.